WetNet: A Spatial–Temporal Ensemble Deep Learning Model for Wetland Classification Using Sentinel-1 and Sentinel-2

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***Abstract*— While deep learning models have been extensively applied to land-use land-cover (LULC) problems, it is still a relatively new and emerging topic for separating and classifying wetland types. On the other hand, ensemble learning has demonstrated promising results in improving and boosting classification accuracy. Accordingly, this study aims to develop a classification system for mapping complex wetland areas by incorporating deep ensemble learning and satellite datasets. To this end, time series of Sentinel-1 dual-polarized Synthetic Aperture Radar (SAR) dataset, alongside Sentinel-2 multispectral imagery (MSI), are used as input data to the model. In order to increase the diversity of the extracted features, the proposed model, herein called WetNet, consists of three different submodels, comprising several recurrent and convolutional layers. Furthermore, multiple ensembling sections are added to different stages of the model to increase the transferability of the model (to other areas) and the reliability of the final results. WetNet is evaluated in a complex wetland area located in Newfoundland, Canada. Experimental results indicate that WetNet outperforms the state-of-the-art deep models (e.g., InceptionResnetV2, InceptionV3, and DenseNet121) in terms of both the classification accuracy and processing time. This makes WetNet an efficient model for large-scale wetland mapping application. The python code of the proposed WetNet model is available at the following link for the sake of reproducibility: https://colab.research.google.com/drive/1pvMOd3_tFYaMYGyHNfxqDxOiwF78lKgN?usp=sharing**

***Index Terms*— Complex land cover, convolutional neural network (CNN), ensemble learning, recurrent neural network (RNN), Sentinel imagery, time series.**

I. Introduction

WETLANDS are known as one of the most important and valuable natural resources on earth. They significantly affect the biological, ecological, and hydrological processes at both regional and global scales [1]–[3]. Wetlands are valuable environmental resources and provide several services to the ecosystem, including water storage and purification, shoreline protection, and carbon sequestration, and provide food and shelter to a variety of plants and animal species [4]. Despite these benefits, wetlands are increasingly being degraded, mainly because of anthropogenic activities, such as industrial development, agricultural intensification, and deforestation, as well as natural processes and environmental changes, such as global warming [3], [5]. This necessitates long-term monitoring systems and programs to evaluate wetlands' status and preserve them from additional damages [4].

One traditional approach for wetland mapping and monitoring is holding field campaigns, but this is laborious and costly. In addition, this approach is impractical for inaccessible wetlands [3]. Remote sensing technologies address the limitations of traditional approaches and provide efficient tools for generating land-use land-cover (LULC) maps. In particular, the integration of multisource remote sensing data, such as SAR and multispectral data, provides additional opportunities for monitoring the Earth characteristics and dynamics [4].

Both airborne and spaceborne remote sensing data are useful for mapping and monitoring wetlands depending on the scale of the study area and target accuracy. Previous studies reported the success of producing wetland maps and inventories by such data. These inventories act as baseline datasets for further applications, such as the development of regional, national, and international wetland policies, determining endangered wetland species habitat for restoration or protection, evaluation of wetland services and benefits, and identification of both past and future trends and changes [6].

Generally, in wetland areas, water is present either at the surface, below vegetation canopy, or in the soil, thus making it difficult to attribute a distinct spectral or backscattering signature to the wetland classes [4]. In addition, wetlands could change rapidly because of the dynamics of water level, water area, and vegetation growth [3]. Accordingly, wetland mapping could be a challenging task, and its success depends on the selection of the appropriate tools and data. In particular, multitemporal remote sensing data with a sufficient spatial and temporal resolution are required to increase the reliability of the wetland products.

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The Copernicus program of the European Space Agency (ESA) provides a unique opportunity for wetland mapping and monitoring with its freely available high spatial and temporal resolution data collected by Sentinel-1 Synthetic Aperture Radar (S1 SAR) and Sentinel-2 Multispectral (S2 MS) sensors. S1 consists of a dual-polarized C-band sensor, which is independent of weather and day/night conditions, and has the capability to penetrate through soil and vegetation canopies. Furthermore, such data are sensitive to the targets' geometrical and electrical features, such as surface roughness, dielectric constant, and water content, all of which are important for wetland vegetation mapping. Sentinel-2 satellites (i.e., S2-A and S2-B) provide optical images with their multispectral sensor in the visible, near-infrared (NIR), and short-wave infrared (SWIR) portions of the electromagnetic spectrum. S1 and S2 provide complementary features for wetland monitoring from space. More specifically, multispectral optical data with their nanometer wavelengths are sensitive to the molecular scale and chemical characteristics of targets, while radar signals are responsive to the physical and geometrical structures of targets, including moisture content and surface roughness. Furthermore, radar's long wavelength is capable of penetrating through the surface. In particular, HV polarization is sensitive to volume scattering and canopy structure, whereas HH polarization has demonstrated high sensitivity to vegetation flooding, which is useful for wetland mapping [3], [4], [7]. Both missions utilize two polar-orbiting satellites with revisit times of, respectively, five and six days for S-1 and -2 at the equator, providing data with a relatively high temporal resolution for wetland mapping [4].

Recently, several studies used S1 and S2 data for wetland mapping. Mahdianpari *et al.* [8] generated the wetland inventory map of Newfoundland with an approximate area of 106 000 km^2 in Canada by combining the multitemporal S1 and S2 images at a spatial resolution of 10 m. Slagter *et al.* [4] used 65 dual-polarized S1 and 286 S2 images in the period of 2016 to 2018 for mapping the St. Lucia wetlands located at the eastern shore of South Africa. Dang *et al.* [9] mapped the coastal wetland of an area in Vietnam and monitored changes within these wetlands by combining S2 multitemporal images with ALOS PALSAR data and NOAA digital elevation models (DEMs).

Handling and accurate processing of the high dimensional remote sensing data can be challenging using traditional remote sensing approaches, yet the nonparametric machine learning (ML) methods, such as decision tree (DT), support vector machine (SVM), and random forest (RF), make this task much easier and feasible. Compared to the traditional parametric classifiers, such as a maximum likelihood or Bayesian classifiers, nonparametric models do not consider any preassumption about the probability distribution function (pdf) of data. This is particularly important when combining multisource data as input to the classifier. Furthermore, the ability to handle a large number of input features, subspace mapping, and supervised or unsupervised classification strategies is commonly used techniques in the advanced ML methods for LULC mapping.

Many studies demonstrate the excellence of ML methods for LULC classification in comparison with the classic methods [9], [10]. However, the quality of the results still has a high dependence on the input features [12]. Therefore, similar to the classical methods, proper feature selection is a crucial step in ML approaches.

Most recently, with the advent of powerful processors and GPUs, several studies used deep learning (DL) methods for solving these complex problems. DL models, including feed-forward multilayer perceptron (MLP), convolutional neural network (CNN), and recurrent neural network (RNN), have obtained state-of-the-art results in various remote sensing problems by providing innovative and flexible model architectures. Basically, DL methods, in addition to being a classifier, are able to automatically extract optimum features from the input data.

CNNs consist of multiple adaptive filters and have been used for processing remote sensing imagery in different applications and addressing computer vision problems. There are two general approaches for image classification with CNNs, namely patch-based CNN and a fully convolutional network (FCN). DeLancey *et al.* [13] in a study area in Alberta investigated the effectiveness of CNN models for classifying wetland areas. According to their results, CNN models boosted the discriminability of wetland classes more than 10% compared to the other traditional ML methods. Rezaee *et al.* [14] classified wetland classes by using three features of RapidEye multispectral image as the input data to the AlexNet model and trained the model with a fine-tuning technique. Mahdianpari *et al.* [15] evaluated the capability of well-known CNN architectures using the patch-based method for wetland mapping using high-resolution multispectral data. According to their results, InceptionResNetV2, ResNet50, and Xception models obtained the best results. In another study, Mohammadimanesh *et al.* [7] proposed an FCN model for Wetland mapping using the Radarsat-2 Polarimetric dataset and demonstrated the effectiveness of Inception and residual blocks for classifying wetland areas. U-net FCN was used in [9] for coastal wetland mapping with S2 images. Sun *et al.* [16] used SegNet FCN architecture for classifying croplands using a Landsat dataset.

Apart from CNN architectures, RNNs are popular, given their ability to extract features from sequential data. Minh *et al.* [17] used S1 time series and RNNs for classifying vegetation quality in the winter. In a similar approach, Ndikumana *et al.* [18] evaluated multilayer deep RNNs, including long short-term memory (LSTM) and gated recurrent unit (GRU), for classifying agricultural lands using a multitemporal S1 dataset. Based on their results, five-layered LSTM obtained the best result for their study area. Sharma *et al.* [19] proposed a patch-based RNN model for spatial, spectral, and temporal classifications of Landsat multispectral datasets.

DL methods, due to the stochastic nature of their training process, may result in a different model after each training process. Therefore, recent studies combined ensemble methods with DL in order to improve the robustness and accuracy of the results and reduce the stochastic manner of the DL-based

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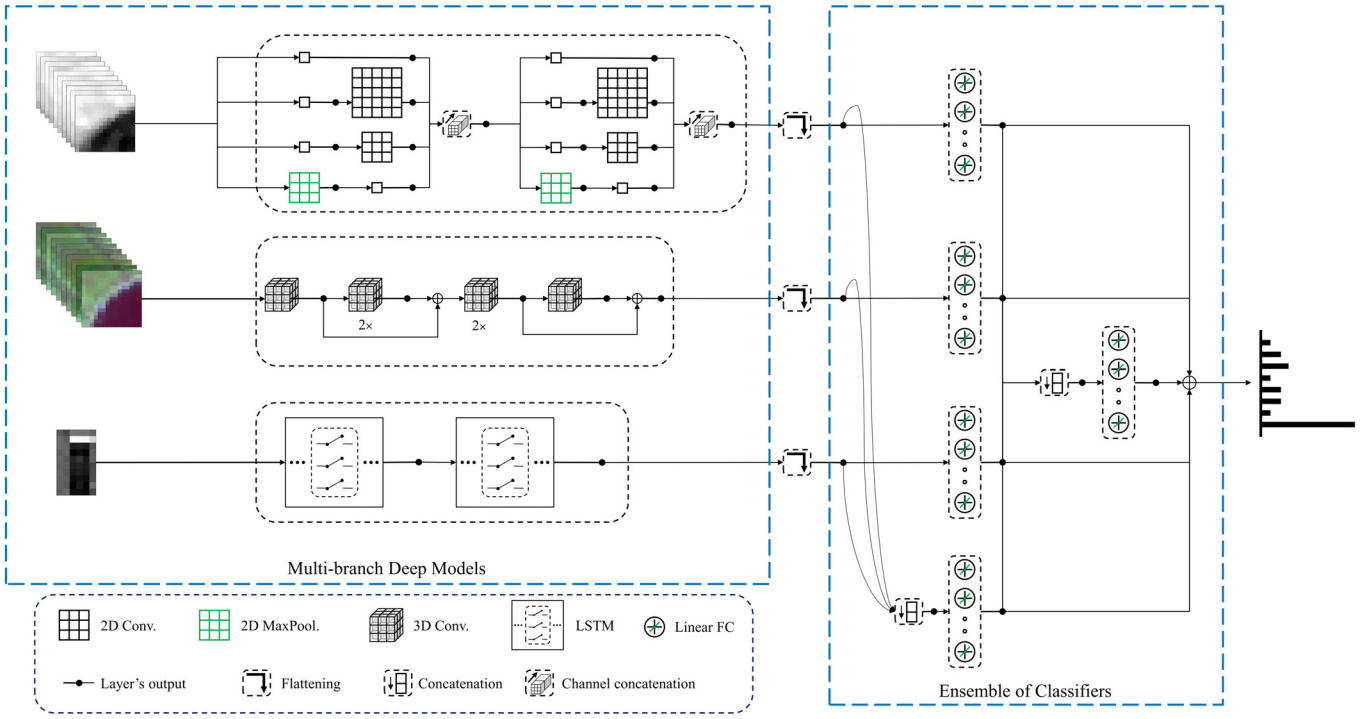

Fig. 1. Overview of the proposed model, WetNet, for classifying wetland areas using S1 and S2 datasets.

methods. Chen *et al.* [20] proposed a deep ensemble model for hyperspectral image classification by combining a random feature selection method with the ensemble of three CNN models. The final classification label was selected based on the majority voting of the individual classifier's output. Furthermore, they proposed a transfer learning approach from a single trained model to the ensemble models to reduce the processing time. Fan *et al.* [21] proposed an ensemble model by combining three well-known CNNs, namely ResNet50, ResNeXt, and ShuffleNet-V2, for classifying the urban area using submeter resolution images in a semisupervised approach. The final label was selected based on the maximum probability acquired by averaging the ensemble models' outputs.

This article aims to improve the classification accuracy of wetland complexes by proposing a novel deep ensemble model specifically designed to capture important wetland characteristics. This new method leverages both convolutional and recurrent networks to exploit all embedded temporal, spectral, and spatial information from multisource remote sensing data. In this model, each convolutional and recurrent section is trained simultaneously in different branches, and the final label is determined using the majority voting between the outputs of all branches. To the best of our knowledge, this is the first deep ensemble method, proposed for classifying Canadian wetlands, and attained state-of-the-art classification accuracy for these spectrally similar wetland classes.

II. WetNet Structure

This section explains the details of the proposed WetNet model for wetland classification. As shown in Fig. 1, the proposed model consists of two major parts, multibranch deep layers, and multiple classifier ensembles. Each part has input data and comprises a deep submodel. The output features of each submodel are connected to a classifier, and the final classification score is determined based on the ensembling of the classifiers' output. Sections II-A–II-F describe the proposed WetNet in more detail.

A. Ensemble Model

Ensemble models prevent model overfitting and yield better generalization capability [22]. It is basically constructed based on multiple base classifiers and then combining their results through, for example, majority voting [23]. This approach reduces the accuracy deviations made by an individual classifier and improves the robustness and reliability of the classifier [20], [22].

In ensemble learning, it is generally recommended to use heterogeneous base classifiers, as the higher diversity between them results in a better generalization [20], [23]. In DL, there exist many models, each with a different perception capability and is designed to extract different kinds of features from the data. Accordingly, the combination of different DL models achieves a better generalized perception of the data and, thus, improves the results.

In this study, three types of DL models are implemented as submodels in three parallel branches for extracting deep features from three types of input data. The first two submodels (branches) use deep 2-D and 3-D convolutional layers, respectively, to extract deep features from a time series of the multipolarized S1 and multispectral S2 datasets. The third branch exploits recurrent layers in order to exclusively extract

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temporal features from the input multivariate and multitemporal data. At the end of each deep branch, a fully connected (FC) layer is deployed to estimate the labels' score as a miniclassifier. In addition, a fourth miniclassifier is implemented to use the concatenated deep features of all branches. Finally, the outputs of these four miniclassifiers are concatenated as a feature vector to be fed to another FC layer for determining the labels' score (i.e., a metaclassifier). Notably, the proposed model produces five classification outputs, and the final label is based on the majority voting between the aforementioned outputs.

B. 2-D CNN Submodel

The first branch of the model utilizes 2-D convolutional layers to process the input from S1 image patches. $x_1(r, c, d)$ is considered as an element of the 3-D tensor of the input S1 patch with a size of $R \times C \times D_1$, where the width and height of the input tensor are denoted as R and C, respectively. D_1 is the dimension of the stacked dual-polarized and temporal S1 dataset.

In a 2-D convolutional layer, the input tensor is fed to a bank of learnable filters, consisting of N 2-D kernels of size $U \times V$, where the output is the summation of all input in the d dimension. Therefore, convolving the nth 2-D kernel $K_{2D}(u,v,n)$ with a 3-D input tensor results in a new tensor, known as feature map, $x_{fm}(r', c', n)$, with a size of $R' \times C' \times N$, which can be calculated as

$$x_{fm}(r', c', n) = \sum_{d=1}^{D} \sum_{u=0}^{U-1} \sum_{v=0}^{V-1} x_1(r-u, c-v) K_{2D}(u, v, n) \quad (1)$$

where $x_{fm}(r', c', n)$ is an element of the resulting feature map. It is worth noting that, if the input tensors are padded, the width and height of the resulting feature map are the same as the input tensor. In order to increase the nonlinearity, the output can be fed to a nonlinear function f, which, in this case, is the rectified linear unit (ReLU) function

$$f(x_{fm}(r', c', n)) = \max\{0, x_{fm}(r', c', n)\}. \quad (2)$$

Also, batch-normalization (BN) is applied to rescale the outputs, which results in a better and faster learning procedure [24].

The implemented architecture in this submodel is adopted from the Inception network [25], where two consecutive Inception modules with dimension reduction are exploited. Inception module using filters with varying sizes at the same layer extracts more detailed features with different sizes while adding few parameters and overcomes overfitting and computational cost problems. One key point is using a 1×1 convolutional layer (i.e., a bottleneck layer) at the first stage before larger filters, which makes the network deeper and adds more nonlinearity to the architecture [15], [26]. The efficiency of such an inception block for wetland classification has been previously confirmed by Mohammadimanesh *et al.* [7] who demonstrated the effects of using multiscale filters of the inception network for extracting useful wetland features.

Each implemented inception module contains two stages. At the first stage, the input tensor is simultaneously fed to three parallel 1×1 convolutional layers and one 3×3 max-pooling layer. Two of the mentioned convolutional layers are fed to 1×1 and 1×1 convolutional layers, respectively. In addition, the output of the pooling layer is fed to another 1×1 convolutional layer, followed by another convolutional layer. The final resulting feature maps are concatenated in three dimensions in order to generate a deeper feature map. Therefore, layers' output can be formulated as follows:

$$x_{fm,1}^{(i)}(r, c, n) = \max\left\{0, \sum_{d=1}^{N_0} x_{fm}^{(i-1)}(r, c, n) K_{2D}(1, 1, n)\right\} \quad (3)$$

$$x_{fm,3}^{(i)}(r, c, n) = \max\left\{0, \sum_{d=1}^{N_0} \sum_{u=0}^{2} \sum_{v=0}^{2} x_{fm}^{(i-1)}(r-u, c-v) K_{2D}(u, v, n)\right\} \quad (4)$$

$$x_{fm,5}^{(i)}(r, c, n) = \max\left\{0, \sum_{d=1}^{N_0} \sum_{u=0}^{4} \sum_{v=0}^{4} x_{fm}^{(i-1)}(r-u, c-v) K_{2D}(u, v, n)\right\} \quad (5)$$

$$x_{fm,\max \text{Pool2D}}^{(i)}(r, c, d) = \max_{u,v \in A}\left\{x_{fm}^{(i-1)}(u, v, d)\right\} \quad (6)$$

$$X_{fm} = S\left\{\left[x_{fm,1}^{(1)}, x_{fm,1}^{(2)}, x_{fm,3}^{(2)}, x_{fm,5}^{(2)}\right]\right\} \quad (7)$$

where $x_{fm,1}^{(i)}(r, c, n)$, $x_{fm,3}^{(i)}(r, c, n)$, and $x_{fm,5}^{(i)}(r, c, n)$ are elements of the resulting feature maps by the nth 1×1, 3×3, and 5×5 filters in the ith layer, followed by the ReLU function. $x_{fm,\max \text{Pool2D}}^{(i)}(r, c, d)$ denotes the result of the 2-D max-pooling operator for an element of the dth channel and A is the area of the max-pooling operator, and in this case, $A = 3 \times 3$. Finally, the output of the inception module is denoted by X_{fm} where $S\{\cdot\}$ is the channel stacking operator.

C. 3-D CNN Submodel

WetNet has a branch of multiple 3-D convolutions for processing the S2 multitemporal multispectral input patches. 2-D kernels, as described in (1), may overlook some spectral–temporal features because of the summation operator in the third dimension of the input tensor (d), while the proposed model aims to exploit multispectral data. This drawback is addressed using 3-D kernels that extract the embedded relations within all three dimensions of the input tensor.

Therefore, $x_2(r, c, d)$ is considered an element of the 3-D input tensor with a size of $R \times C \times D_2$, where D_2 is the dimension of the stacked multispectral and multitemporal S2 dataset. Accordingly, given $K_{3D}(u,v,w,n)$ as a 3-D kernel of a bank of learnable filters comprised of N 3-D kernels of size $U \times V \times W$, the resulting output will be N feature maps; each can be calculated as (3)

$$x_{fm}(r', c', d', n) = \sum_{u=0}^{U-1} \sum_{v=0}^{V-1} \sum_{w=0}^{W-1} x_2(r-u, c-v, d-w) K_{3D}(u, v, w, n) \quad (8)$$

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where $x_{fm}(r', c', d', n)$ is an element of the resulting nth feature map.

Adding multiple layers to the CNN architecture makes it deeper; however, training of the model may face problems because of vanishing gradient problem in the backpropagation process, which results in a decreased performance (i.e., the degradation problem) [20], [26], [27]. The residual learning approach, which employs deep residual learning modules via additive identity transformations, has been proposed to overcome the vanishing gradient problem [27]. Specifically, the residual module uses a direct path between the input and the output, and each stacked layer fits a residual mapping rather than directly fitting a desired underlying mapping [27]. Notably, the optimization is much easier on the residual map than the original, unreferenced map while adding neither an extra parameter nor computational complexity [15]. Therefore, the output of the 3-D residual feature map is formulated as follows:

$$\begin{aligned} &x^{(i)}_{\text{res}-fm}(r', c', d', n) \\ &= x^{(i-1)}_{fm}(r, c, d) \\ &\quad + \sum_{u=0}^{U-1}\sum_{v=0}^{V-1}\sum_{w=0}^{W-1} x^{(i-1)}_{fm}(r-u, c-v, d-w) K_{3D}(u, v, w, n) \end{aligned} \tag{9}$$

where $x^{(i-1)}_{fm}(r,c,d)$ is an element of a feature map from the previous layer and $x^{(i)}_{\text{res}-fm}(r',c', d',n)$ is an element of the resulting nth feature map of the residual block.

D. RNN Submodel

The third branch of WetNet is comprised of recurrent layers, where the input is a 2-D array containing the multivariate temporal information. This part of the network exclusively aims to extract temporal relations from the input data by stacking the RNN layers. In contrast to the mentioned feedforward layers, recurrent layers are able to explore and extract the relations within the sequential data, by using a recurrent hidden state.

If $x = [x_1, x_2, \ldots, x_T]^T$ is denoted as the input sequence in the time series of T, where x_i is data at the ith time step, a simple recurrent unit updates its recurrent hidden state h_t at time step t by

$$h_t(x_t) = H(\Theta_{hh} h_{t-1}(x_{t-1}) + \Theta_{hx} x_t + \theta_h) \tag{10}$$

where θ_h is the bias term and H denotes the nonlinearity function. Θ_{hh} and Θ_{hx} are the weight matrices that connect the hidden state from the previous step to the current step and the input data in the current step to the hidden state, respectively [28].

Importantly, a simple RNN is not capable of extracting complex relations. Also, using long sequential data may result in vanishing or exploding gradients during the training process [28], [29]. To address this problem, the LSTM unit was proposed by Hochreiter and Schmidhuber [30]. The LSTM's solution is to use three types of gates to control the forgetting and remembering of information. In particular, given the sequential input data (x_t) and the previous time's hidden state (h_{t-1}), forget gate (f_t) decides which information should be kept in the network, while input gate (i_t) decides what new information should be used, and the output gate (o_t) decides unit's output

$$f_t(x_t) = \sigma\left(\Theta_{fh} h_{t-1}(x_{t-1}) + \Theta_{fx} x_t + \theta_f\right) \tag{11}$$

$$i_t(x_t) = \sigma(\Theta_{ih} h_{t-1}(x_{t-1}) + \Theta_{ix} x_t + \theta_i) \tag{12}$$

$$o_t(x_t) = \sigma(\Theta_{oh} h_{t-1}(x_{t-1}) + \Theta_{ox} x_t + \theta_o). \tag{13}$$

These gates use a sigmoid function (σ) to decide which portion of the information should be passed in a soft manner with a value ranging between 0 and 1. If Sigmoid's output is 1, all the information is passed to the gate, and if the output is 0, no information is passed [30], [31]. According to the values of forget and input gates, cell state (C_t) is updated as follows:

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \tag{14}$$

where $\tilde{C}_t$ is the candidate cell information that is calculated inside the current LSTM unit:

$$\tilde{C}_t = \tanh(\Theta_{ch} h_{t-1}(x_{t-1}) + \Theta_{cx} x_t + \theta_c). \tag{15}$$

Finally, the hidden recurrent state (h_t) is updated based on the output gate and the cell state

$$h_t(x_t) = o_t \times \tanh(C_t). \tag{16}$$

E. Objective Function

The proposed ensemble model uses three submodels that are implemented in three parallel branches. The final layer of each submodel consists of an FC layer, which results in a classification score for each submodel. Therefore, rather than separately training each submodel, the whole model is trained simultaneously within an end-to-end processing scheme. Each classifier has its own objective (i.e., loss) function. The model is trained by backpropagating the obtained error value at each epoch from each objective function. Therefore, the final objective function (l) can be formulated as follows:

$$\begin{aligned} &l(X; \psi_e, \psi_b, \psi_d) \\ &= \gamma_e l_e(X; \psi_e) + \gamma_d l_d(X; \psi_d) + \sum_{m=1}^{3} \gamma_{b,m} l_{b,m}(X; \psi_{b,m}). \end{aligned} \tag{17}$$

As indicated in (17), the proposed model comprises three loss functions, namely, submodels (l_b), a loss function based on ensembling features of submodels (l_e), and a decision loss function (l_d), which is obtained based on the final FC layer of the proposed model. $X = \{x;y\}$ is set of training samples (x) and their target values (y). ψ denotes the estimated classification score from each corresponding FC layer. γ represents the weight value of each corresponding loss function, and it is set to one in this study.

This study uses the squared-hinge loss function with the L2 regularization term for all the classifiers [32]. The hinge loss has been extensively used in SVM and has shown strong capabilities in reducing the overfitting problem. The square operator makes it derivable for backpropagating the error [32]–[34]. Therefore, the utilized loss function can be formulated as follows:

$$l(X; \psi) = (1 - \psi^T y)^2 + \lambda \|\Theta\|_2^2 \tag{18}$$

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TABLE I
WETNET CONFIGURATIONS OF EACH LAYER AND SUBMODEL

	Sub-model 1	**Sub-model 2**	**Sub-model 3**
Inputs [shape]:	S1 [15×15×20]	S2 [15×15×40]	S1,S2 [10×6]
Feature extraction layers:	1×1 2D Conv.+ReLU+BN (16) 1×1 2D Conv.+ReLU+BN (64) 1×1 2D Conv.+ReLU+BN (96) 1×1 2D MaxPool 5×5 2D Conv.+ReLU+BN (32) 3×3 2D Conv.+ReLU+BN (128) 1×1 2D Conv.+ReLU+BN (32) Channel Concat. 1×1 2D Conv.+ReLU+BN (16) 1×1 2D Conv.+ReLU+BN (64) 1×1 2D Conv.+ReLU+BN (96) 1×1 2D MaxPool 5×5 2D Conv.+ReLU+BN (32) 3×3 2D Conv.+ReLU+BN (128) 1×1 2D Conv.+ReLU+BN (32) Channel Concat.	3×3×3 3D Conv.+ReLU+BN (64) [stride: 2] 3×3×3 3D Conv.+ReLU+BN (64) 3×3×3 3D Conv.+ReLU+BN (64) Residual Block 3×3×3 3D Conv.+ReLU+BN (64) 3×3×3 3D Conv.+ReLU+BN (64) 3×3×3 3D Conv.+ReLU+BN (64) [stride: 2] Residual Block	LSTM+tanh (1024) LSTM+tanh (512)
	Flatten1	Flatten2	Flatten3
Classifier	FC + Linear (8) [Classifier1]	FC + Linear (8) [Classifier2]	FC + Linear (8) [Classifier3]
Feature Fusion	Concat [@Flatten1; @Flatten2; @Flatten3]		
Classifier	FC + Linear (8) [Classifier4]		
Classifier Fusion	Concat [@Classifier1; @ Classifier2; @Classifier3; @Classifier4] FC + Linear (8) [Classifier5]		

where Θ is the weights of the corresponding FC layer and λ controls the regularization term.

F. Model Parameters

Table I lists the details of the configuration setting for each submodel of WetNet. The model's parameters are trained based on the minibatch backpropagation algorithm. The error at each epoch is calculated based on the determined objective function, and the parameters are then updated. The adaptive momentum (Adam) optimizer, with the minibatch size of 32 and a learning rate of 0.001 with exponential decay, is used as the optimization algorithm.

III. EXPERIMENTS AND RESULTS

A. Study Area and Dataset

The study area is located in Avalon, in the eastern part of Newfoundland Island, Canada, with an approximate area of 3550 km^2. In general, the Island of Newfoundland has a cool summer and a humid continental climate, which is greatly affected by the Atlantic Ocean. The study area in Avalon is mainly identified by an oceanic climate, foggy, cool summers, and relatively mild winters [35]. Several wetland classes categorized by the Canadian wetland classification system, including bog, fen, marsh, swamp, and shallow water, are found within the study region. Because of the island's climate aiding extensive peatland formation, bog and fen are the dominant wetland classes in the study area. Other land cover classes are upland, deep water, and urban/bare land. The urban and bare land classes, both having either an impervious surface or exposed soil, include bare land, roads, and building facilities and are merged into one class in the final classification map [36].

Ground-truth data were collected from the combination of two field campaigns during 2015–17 [7], [14] and 2019 [37]. The 2015–2017 ground dataset was collected during the summer and fall. The spatial distributions of different wetland types and other land cover classes were determined and recorded along with photographs and field notes to facilitate the wetland boundary delineation. Furthermore, global positioning system (GPS) locations were recorded for all the visited land cover types. Another *in situ* wetland dataset was collected in 2019 by investigating 77 sample sites. These wetlands were then visited in the field over the course of a week in September and October and classified as one of four wetland classes (bog, fen, swamp, or marsh) using the Canadian Wetland Classification System. GPS points, photographs, and field notes were collected to report on dominant vegetation type, hydrology, and adjacent land use in the field. These wetlands were later delineated using the VHR imagery, LiDAR-derived DEM, and the ancillary field data, creating 77 classified wetland polygons. Furthermore, the Crop

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TABLE II
NUMBER OF TRAINING AND TESTING POLYGONS AND SAMPLES FOR EACH CLASS

	Training		Testing	
	Polygons	Samples	Polygons	Samples
Bog	95	1300	47	10000
Fen	29	1300	50	2856
Marsh	47	1300	84	2899
Swamp	46	1300	39	1847
Shallow water	17	1300	31	7461
Urban	95	1300	225	10000
Deep water	7	1300	9	10000
Upland	26	1300	41	10000
Total	**362**	**10400**	**526**	**55063**

Inventory map from Agriculture and Agri-Food Canada was used as a guide for locating particular nonwetland land cover classes and determining the most common nonwetland land cover in the city. This resulted in a total of 395 polygons representing nonwetland land cover across the study area [37].

The final dataset consists of 888 polygons, including 485 bog, fen, swamp, marsh, and shallow-water polygons and 403 nonwetland polygons. The 362 and 526 polygons were then divided as training and testing polygons, respectively. Where a total of 1300 random samples for each class are selected from the training polygons, the testing polygons are comprised of 55 063 samples (see Table II). Fig. 2 depicts the testing and training polygons' distribution over the study area.

The developed model uses time series images of S1 and S2 datasets for classification. As suggested by recent research [38], the multiyear spring composite is advantageous for wetland mapping in Canada's boreal regions. This is because such time series data capture within-year surface water variations. However, in this study, the multiyear summer composite was used, given that the leaf-on season begins in late spring/early summer on the Island of Newfoundland [8]. Accordingly, in this study, summer and fall image composites of 2016–2020 were acquired. Due to the prevailing cloudy and rainy weather conditions in the study area, it was impossible to find appropriate cloud-free optical data. Thus, optimal cloud-free composites of S2 imagery were generated by exploiting Google Earth Engine (GEE) composite function. Dual-polarized VV and VH S1 imageries were acquired for each season composite by averaging the ascending and descending images in GEE. As a result, for each time, VV and VH SAR images from S1 and four spectral bands (red, green, blue, and NIR) of S2 imagery with 10×10 m ground sampling were acquired. Table III summarizes the details of the used satellite imagery used in this study.

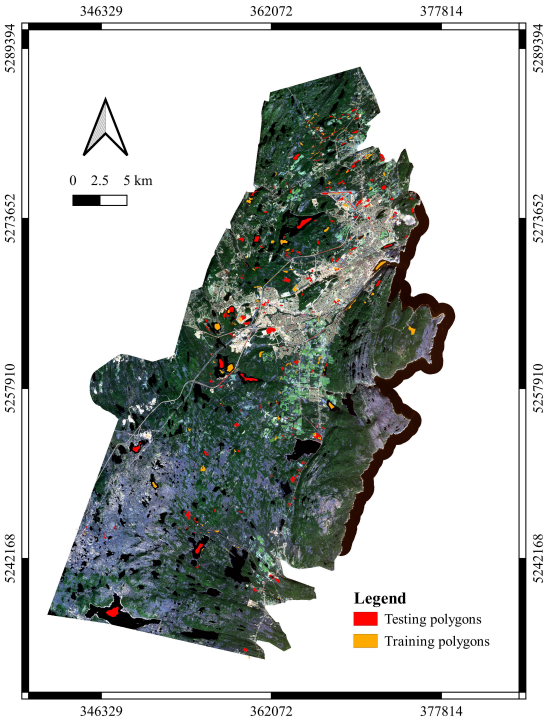

Fig. 2. RGB image of the study area with the distributions of the testing and training polygons.

TABLE III
OVERVIEW OF THE SENTINEL-1 AND -2 DATASETS USED IN THIS STUDY

Data	Features	Cloud cover (%)	Description
S1	σ_{VV} σ_{VH}	-	C-band Level-1 Ground Range Detected (GRD) Sentinel-1 σ_{VV}, σ_{VH}: vertically transmitted, vertically and horizontally received SAR backscattering Coefficients
S2	B2 B3 B4 B8	2016/Summer: 50 2016/Fall: 15 2017/Summer: 1 2017/Fall: 1 2018/Summer: 0.1 2018/Fall: 2 2019/Summer: 0.1 2019/Fall: 0.1 2020/Summer: 0.1 2020/Fall: 10	Sentinel-2 Level-1C data B2: Blue spectral band B3: Green spectral band B4: Red spectral band B8: NIR spectral band

B. Evaluation Metrics

In order to quantify the classification performance of the developed model, two per class metrics, namely, precision (Pr) and recall (Re), as well as three overall metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (k), were employed. These evaluation indices can be represented as follows:

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{19}$$

$$\mathrm{Re} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{20}$$

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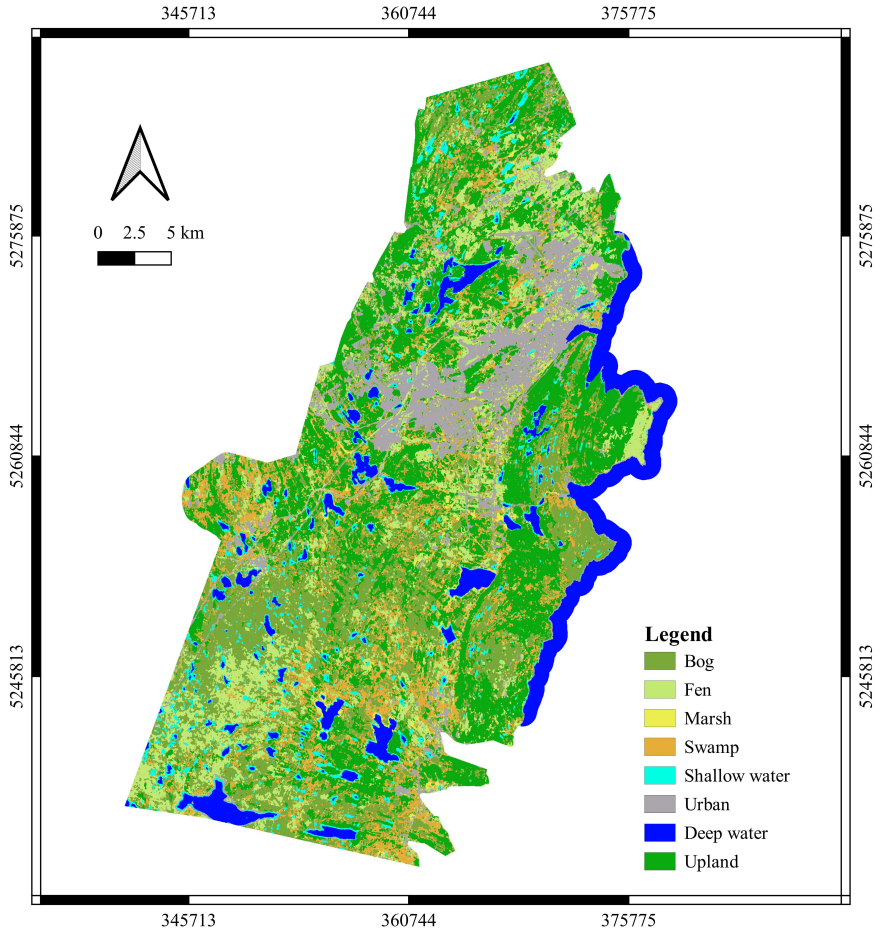

Fig. 3. Classified map of the entire study area.

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} \mathrm{TP}_i}{\sum_{i=1}^{C} (\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{TN}_i + \mathrm{FN}_i)} \tag{21}$$

$$\mathrm{AA} = \frac{1}{C} \sum_{i=1}^{C} \mathrm{Re}_i \tag{22}$$

$$k = \frac{\mathrm{OA} - P_e}{1 - P_e}. \tag{23}$$

C is the number of classes and TP, TN, FP, and FN are true positive, true negative, false positive, and false negative values of each class, respectively. P_e in (23) is random accuracy or chance of agreement and can be calculated as

$$P_e = \frac{\sum_{i=1}^{C} (\mathrm{TP}_i + \mathrm{FN}_i)(\mathrm{TP}_i + \mathrm{TN}_i)}{\sum_{i=1}^{C} (\mathrm{TP}_i + \mathrm{TN}_i + \mathrm{FP}_i + \mathrm{FN}_i)}. \tag{24}$$

C. Classification Results

Fig. 3 shows the classified map of the entire study area, obtained by WetNet. This classification map demonstrates a clear separation of eight land cover classes, namely, bog, fen, marsh, swamp, shallow water, urban, deep water, and upland. To evaluate the efficiency of WetNet, the classification results were compared with several well-known state-of-the-art models, including LSTM [30], VGG16 [39], ResNet50 [27], Inception-ResNet-V2 [40], Inception-V3 [41], DenseNet121 [42], and EfficientNet-B0 [43]. A visual comparison of a small part of the classified maps obtained by the different methods is presented in Fig. 4. LSTM output demonstrates a good performance on linear elements, such as roads or building edges, but it has overestimated the marsh class, especially on the upper right section. On VGG16 results, in some areas, urban is misclassified as shallow water. Furthermore, in some sections, the fen is misclassified as the swamp. ResNet50 and DenseNet121 both demonstrated good results on nonwetland areas; both have overestimated swamps in the whole area and have overestimated bog in the upper left area. In general, EfficientnetB0 obtained appropriate results on nonwetland classes, but there exist some misclassified areas in the central section, where urban structures are misclassified as shallow water. In wetland areas, EfficientnetB0 had good performance on Fen, Marsh, and Swamp, but, in the upper left section, it has overestimated the Bog regions. InceptionResnetV2 and InceptionV3, overall, obtained better visual results compared to other examined models. However, compared to the WetNet, Bog over-estimation in InceptionResNetV2 and Swamp over-estimation in InceptionV3 are evident.

In order to support the presented visual results, Fig. 5 displays the distributions of deep features for each model in 2-D space using the Uniform Manifold Approximation and Projection (UMAP) algorithm [44]. As can be seen,

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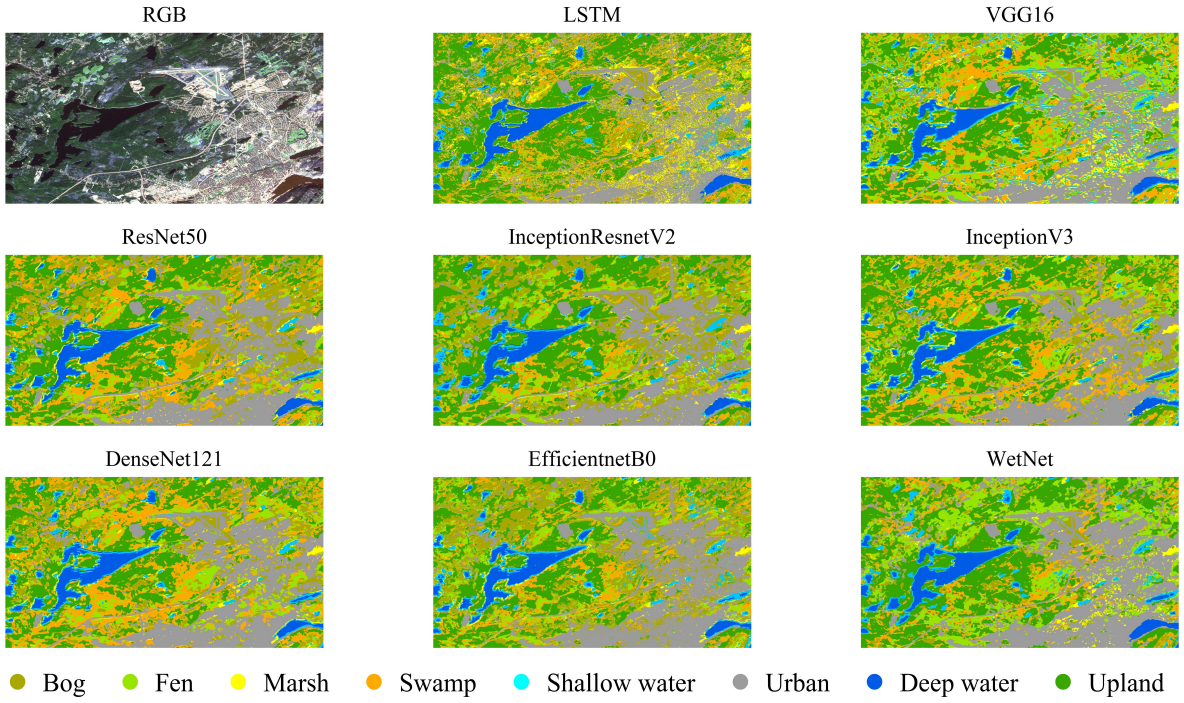

Fig. 4. Classification results obtained from different deep models for a small testing area.

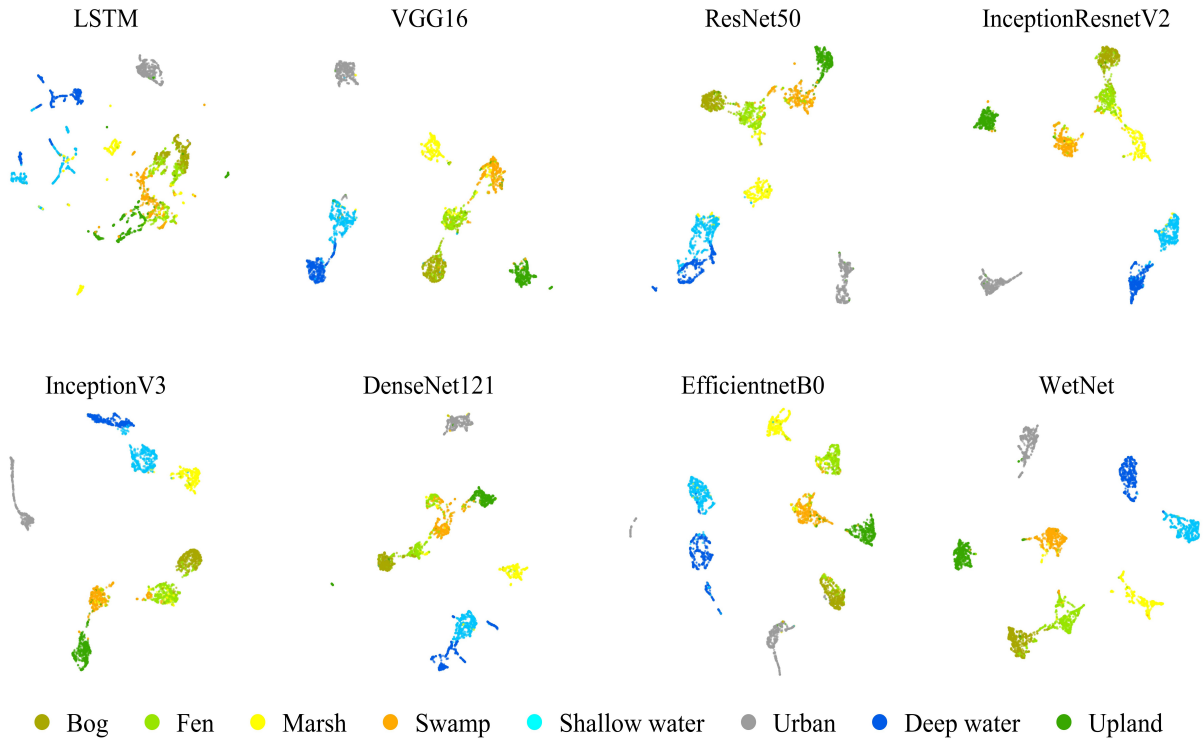

Fig. 5. Feature visualization using the UMAP algorithm for different deep models, as well as the proposed network.

almost all classes represent high separability in urban areas. InceptionResnetV2 demonstrates high discrimination between nonwetland classes. However, WetNet depicts better separation and distributions in wetland classes than the other deep models, especially in Marsh and Swamp samples.

Fig. 6 shows the confusion matrix of all models. As seen, a high number of pixels are located in the diagonal elements of the proposed model. The confusion matrix of the LSTM model represents the effectiveness of extracted temporal features for classifying several wetland classes (e.g., bog and marsh).

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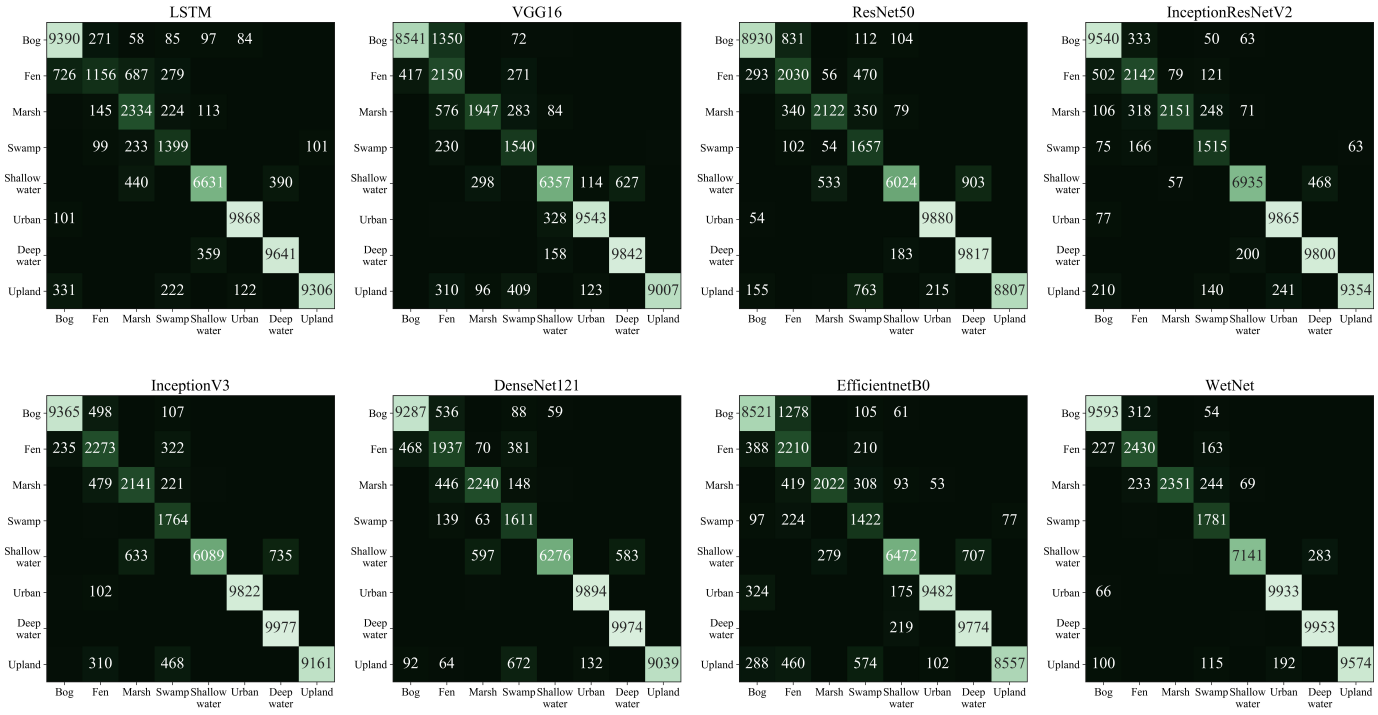

Fig. 6. Confusion matrix of training data based on the prediction of each deep model.

TABLE IV

CLASSIFICATION RESULTS OBTAINED BY VARIOUS DEEP MODELS ON THE TEST DATASET. THE RESULTS OF THE DEVELOPED MODEL ARE SHOWN IN BOLD

		LSTM	VGG16	ResNet50	InceptionResNetV2	InceptionV3	Densenet121	EfficientnetB0	**WetNet**
Bog	Pr:	88.54	94.24	94.63	90.77	**96.53**	93.72	88.54	**96.03**
	Re:	93.90	85.41	89.30	95.40	93.65	92.87	85.21	**95.93**
Fen	Pr:	68.85	46.14	60.67	71.61	61.35	61.41	48.14	**80.20**
	Re:	40.48	75.28	71.08	75.00	79.59	67.82	77.38	**85.08**
Marsh	Pr:	61.75	80.59	74.07	91.30	75.39	73.88	83.90	**96.83**
	Re:	80.51	67.16	73.20	74.20	73.85	77.27	69.75	**81.10**
Swamp	Pr:	63.33	58.87	49.42	73.01	60.95	55.46	54.23	**75.56**
	Re:	75.74	83.38	89.71	82.02	95.51	87.22	76.99	**96.43**
Shallow water	Pr:	92.03	91.00	94.27	94.79	98.88	98.02	92.06	**98.13**
	Re:	88.88	85.20	80.74	92.95	81.61	84.12	86.74	**95.71**
Urban	Pr:	97.96	97.51	97.67	97.47	99.76	98.57	98.08	**97.97**
	Re:	98.68	95.43	98.80	98.65	98.22	98.94	94.82	**99.33**
Deep water	Pr:	95.71	93.90	91.58	95.42	93.03	94.48	93.25	**97.24**
	Re:	96.41	98.42	98.17	98.00	99.77	99.74	97.74	**99.53**
Upland	Pr:	98.60	99.48	99.67	99.26	99.64	99.71	99.09	**99.67**
	Re:	93.06	90.07	88.07	93.54	91.61	90.39	85.57	**95.74**
k (100)		88.64	87.01	87.71	91.97	90.48	89.78	86.05	**95.03**
OA		90.31	88.86	89.47	93.17	91.88	91.27	88.01	**95.81**
AA		83.46	85.04	86.13	88.72	89.22	87.29	84.27	**93.61**
Parameters (M)		3.47	14.948	24.569	55.139	22.867	7.617	4.076	**9.668**
Training time (s/epoch)		2.28	50	74	195	84	79	14	**13**
Prediction time (ms/sample)		0.1115	0.5633	0.4164	1.9399	0.9394	1.0886	0.5865	**0.6637**

Table IV provides a detailed quantitative comparison of the classification results obtained from the proposed methods relative to other deep models.

As seen, the proposed ensemble network outperforms other deep models in terms of both overall and class-based accuracies. As expected, the inception-based classification models

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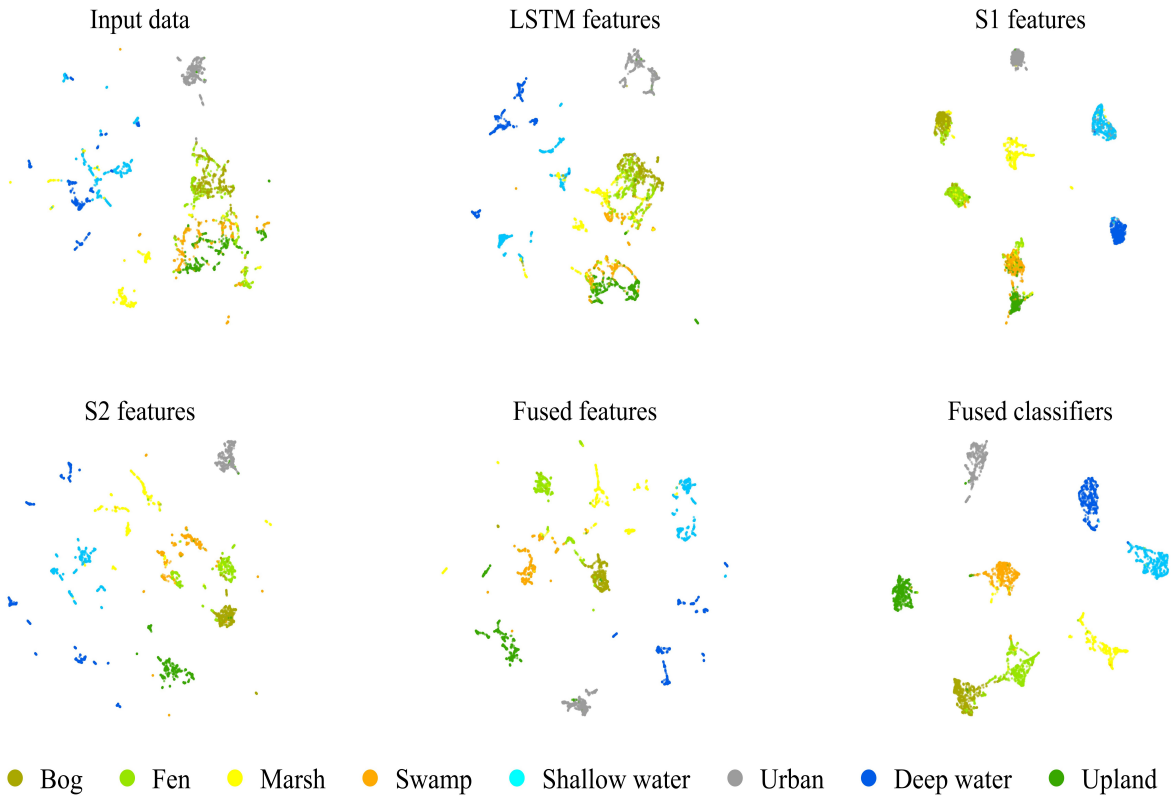

Fig. 7. Feature visualization using the UMAP algorithm for the different layers of the proposed network.

(i.e., Inception-ResNet-V2 and Inception-V3) obtained accuracies comparable to that of WetNet, as all the leveraged multiscale filters are suitable for wetland discrimination. However, the main drawback of Inception-ResNet-V2 and Inception-V3 is that their efficiency depends on training many parameters compared to the proposed method, albeit with lower accuracy. This demands a large number of training samples to reach a target accuracy, which can be problematic, given the difficulty of accessing many wetlands in real cases. As shown in Table IV, training these two models is much slower than other models, given their deep layers and a high number of parameters. Overall, the results presented in Table IV confirmed the efficiency of the proposed model in terms of accuracy, number of training parameters, and training time. This is because the proposed model takes advantage of the most well-known state-of-the-art deep models, albeit with less complexity.

D. Ablation Studies

Ablation studies were conducted in order to evaluate the effectiveness of each submodel, feature fusion, and classifier fusion layer. Table V shows the classification results by removing different branches and layers of the proposed network. As illustrated, the S2 branch is the most effective part of the network, and removing this part of the network decreases accuracy by about 5%∼7% in all metrics. This also confirmed the superiority of optical data compared to SAR imagery for wetland mapping and the efficiency of 3-D residual CNN layers for discriminating small and large wetland classes.

TABLE V
ABLATION ANALYSIS ON THE IMPORTANCE OF EACH SUBMODEL (BRANCH) AND CLASSIFIER FUSION LAYER

	No S1 branch	No S2 branch	No RNN branch	No classifier fusion	**WetNet**
k	93.68	89.35	94.45	92.99	**95.03**
OA	94.67	90.98	95.32	94.09	**95.81**
AA	92.29	87.52	92.97	91.41	**93.61**
WL AA	89.30	83.61	89.83	90.44	**90.87**

In contrast, removing S1 and recurrent branches is less influential on the accuracy indices. Removing the classifier fusion layer caused a 2% decrease in Kappa value, thus indicating the effect of this layer on decreasing the randomness of the results.

Fig. 7 depicts the distribution of the classes at each layer of the network, using the UMAP algorithm representation. As shown, the input samples demonstrate a high degree of overlaps, especially between wetland classes, while the deep submodels are able to partially increase the discrimination between different classes. The separation between classes exceeds within the feature fusion layer, and the classifier layer maximizes separability between samples and represents the least overlaps between samples compared to the previous layers.

Fig. 8 illustrates the neurons' contribution in two concatenation sections of the model, wherein the feature fusion section

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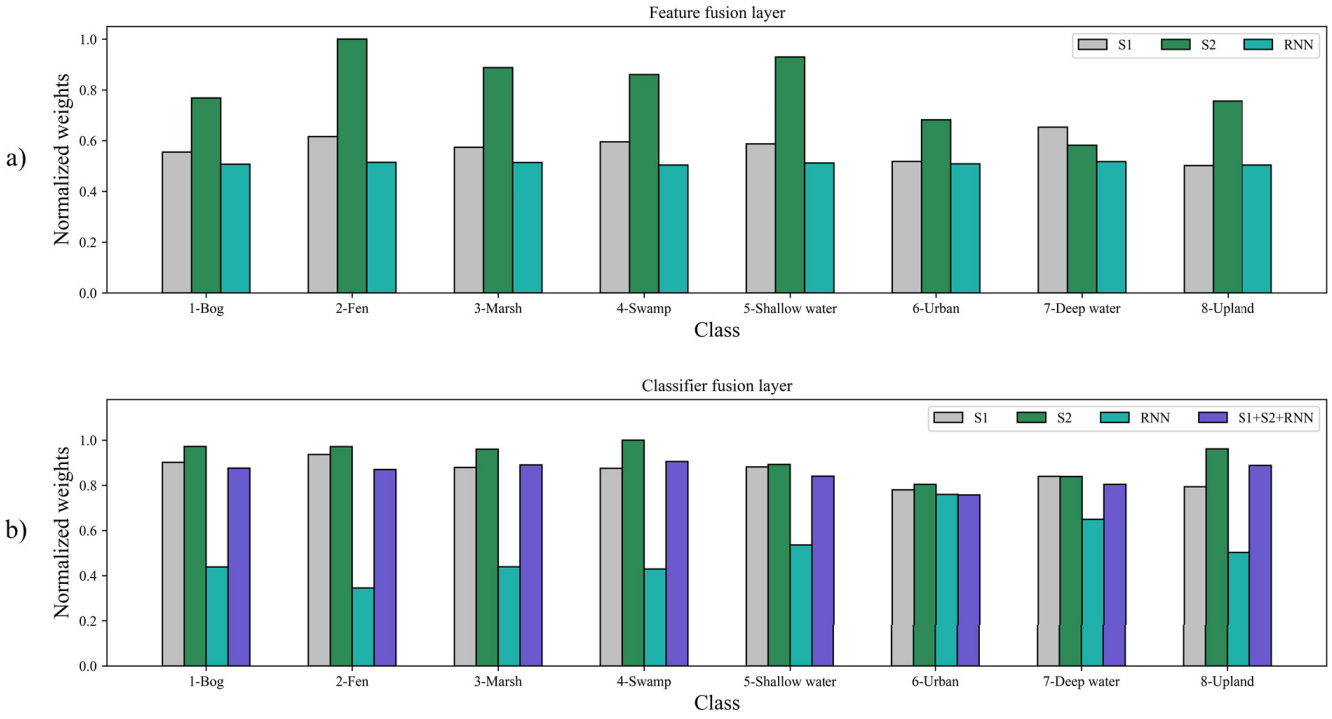

Fig. 8. Neurons' activation across the fusion layers of the proposed network. (a) Normalized weights of the neurons related to each branch of the network after the feature concatenation. (b) Normalized weights of the neurons related to each classifier in the classifier fusion layer.

has three modalities of features (i.e., S1, S2, and multivariate temporal are fused) [45]. In the second concatenation section, the classification scores of four classifiers are concatenated (i.e., classifier fusion). As demonstrated, all input features from each modality (i.e., submodel) demonstrate a high contribution to the next layer's neuron. However, in general, the acquired features from the S2 data contributed more compared to other branches, with a few exceptions. For example, the S1 branch has a slightly higher contribution for deep water, and all modalities have almost equal contributions to the urban class.

IV. Conclusion

This study presented a DL-based model for classifying wetland areas using S1 and S2 multitemporal data. The proposed model, known as WetNet, contains three parallel branches (submodels) of deep models with the ability to ingest three types of input data. The first two branches utilize 2-D and 3-D CNN models to extract deep and nonlinear relations from S1 and S2 input data. The third branch is the LSTM layer and is exclusively designed to extract nonlinear deep temporal features from a multivariate multitemporal array of S1 and S2 data. The extracted features from the three branches are concatenated to generate a broader and more diverse feature space. In addition, each type of features and the fused features are fed to a separate classifier. The outputs of the classifiers are then concatenated and passed to the final classifier. Finally, the class labels are obtained based on the majority voting of the mentioned classifiers.

The efficiency of the proposed model was evaluated in a study area located in Newfoundland, Canada, containing complex wetland classes. The qualitative and quantitative results of the proposed model were compared with seven well-known state-of-the-art deep models. The proposed model achieved an OA of 0.9581 and a Kappa coefficient of 0.9503, which were higher than those of other models. In terms of wetland class-based accuracy, WetNet outperformed all well-known deep models, while it is comprised of a moderate number of trainable parameters and thus, required a shorter processing time relative to other methods. An in-depth evaluation of different layers of the proposed model using various visual and quantitative experiments was conducted. According to the results of these experiments, extracting multimodal features from multimodal datasets resulted in more than 2% improvement in the overall classification results.

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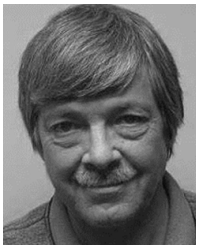

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